EE2001 Electrical Circuits-I:

Techniques of Circuit Analysis

23 October 2023

Outline



1. **Introduction to the Node-Voltage Method**
2. **Introduction to the Mesh-Current Method**

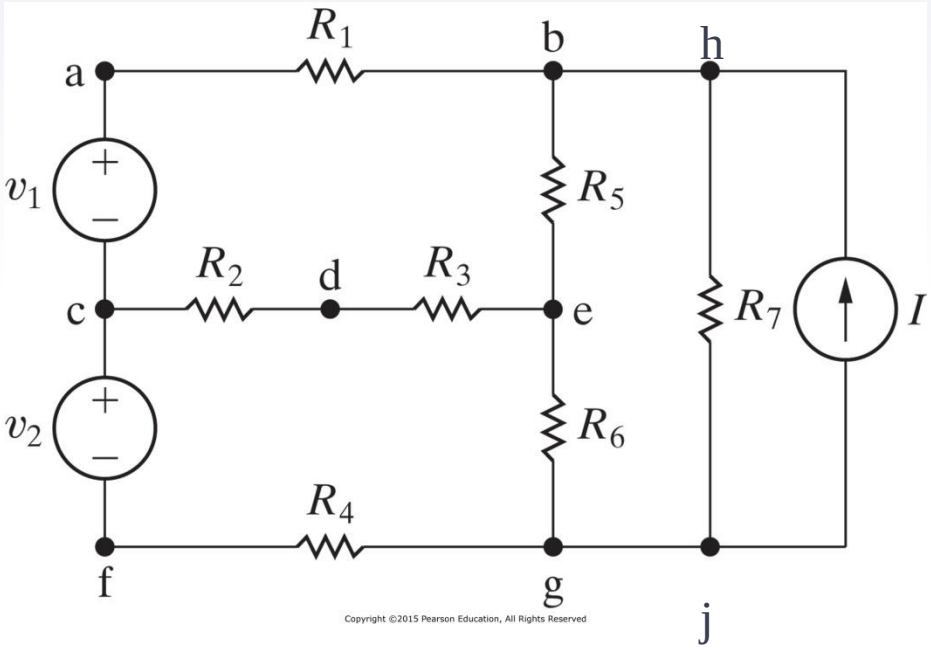
Terminology

1

- A **node** is a point where two or more circuit elements meet (i.e., are connected).
- An **essential node** is a node where three or more circuit elements join (i.e., are connected).
- A **path** is a trace of adjoint basic circuit elements with no elements included more than once.
- A **branch** is a path that connects two nodes.
- An **essential branch** is a path which connects two essential nodes without passing through an essential node.
- A **loop** is a path whose last node is the same as the starting node.
- A **mesh** is a loop that does not enclose any other loop.
- A **planar** circuit is a circuit that can be drawn on a plane with no crossing branches.
- A circuit drawn with crossing branches which can be redrawn with no crossing branches is still a planar circuit.

Terminology

- An example:



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Identify,

a) all nodes

a, b (or h), c, d, e, f and g (or j).

b) all essential nodes

b (or h), c, e and g (or j).

c) all branches

v_1, v_2, R_1, R_2, R_3, R_4, R_5, R_6, $R_7$and I

d) all essential branches

$v_1 - R_1$, $R_2 - R_3$, $v_2 - R_4$, R_5 R_6, $R_7$and I

e) the meshes

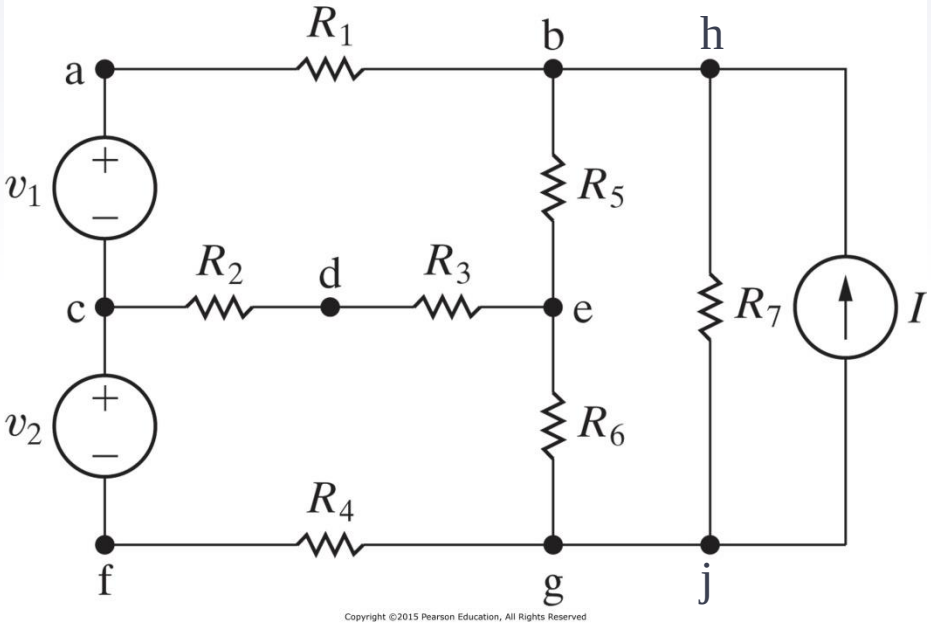
$v_1 - R_1 - R_5 - R_3 - R_2$,

$v_2 - R_2 - R_3 - R_6 - R_4$,

$R_5 - R_7 - R_6$ and $R_7 - I$

Terminology

- An example:



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Identify,

f) two paths that are not loops or essential branches

$R_1 - R_5 - R_6,$

$v_2 - R_2$

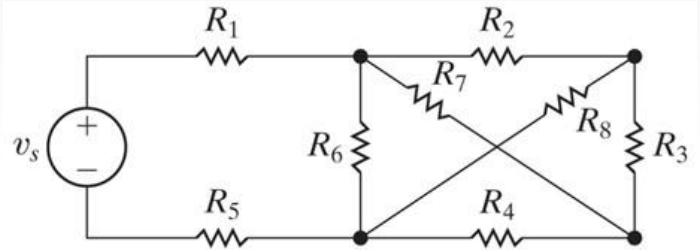
g) two loops that are not meshes

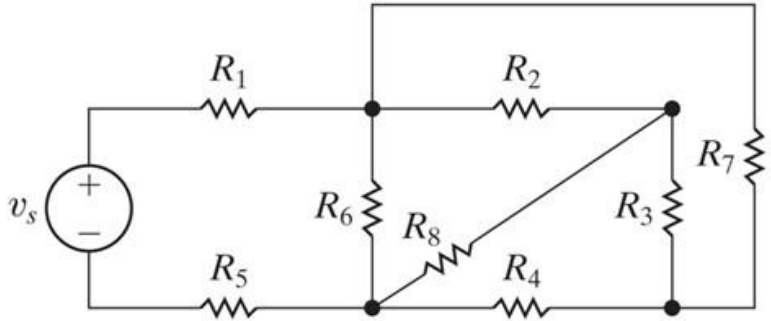
$v_1 - R_1 - R_5 - R_6 - R_4 - v_2,$

$\mathrm{I} - R_5 - R_6$

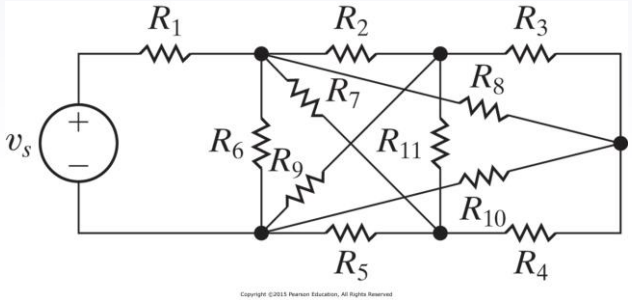
Terminology

- A **planar** circuit





- A **non-planar** circuit



- The **node-voltage method** is applicable to **both planar and non-planar** circuits.
- The **mesh-current method** is limited to **planar** circuits.

Terminology

1 Simultaneous Equations: How Many?

- In a given circuit,
  - The number of **currents to be determined** = the **number of branches**
  - Denote with $\boldsymbol{b}$ the number of **branches with unknown currents**
    - ✓ Then we need $\boldsymbol{b}$ **independent equat**ions to solve the circuit
  - Denote with $\boldsymbol{n}$ the **number of nodes** in the circuit
    - ✓ It is possible to write $(\boldsymbol{n-1})$ **independent KCL equat**ions. The **n-th equation is always dependent on the previous** $(n-1)$ equations.
  - One needs to apply $[\mathbf{b}-(\boldsymbol{n-1})]$ **KVL to the meshes (loops)** to obtain total of b equations
  - Basically,
    - ✓ $(\boldsymbol{n-1})$ **KCL** equations at the **nodes**
    - ✓ $[\mathbf{b}-(\boldsymbol{n-1})]$ **KVL** equations at the **loops or meshes**
    - ✓ Total of $\boldsymbol{b}$ **independent equations** for the **b unknown branch currents**

Terminology

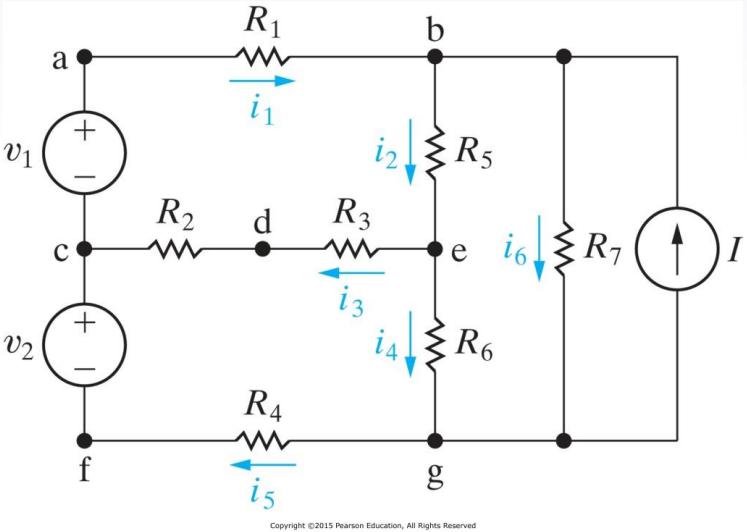
Simultaneous Equations: How Many?

1

- The same relation is valid for the **essential nodes** and **essential branches** with **less number of** equations.
  - Denote with $\boldsymbol{b_e}$ the number of **essential branches (and also the number of unknown currents to be calculated)**
  - Denote with $\boldsymbol{n_e}$the **number of essential nodes** in the circuit
  - It is possible to write
    - ✓ $(\boldsymbol{n_e - 1})$ **KCL** equations at the **essential nodes**
    - ✓ $\boldsymbol{b_e - (n_e - 1)}$ **KVL** equations at the **loops or meshes**
    - ✓ Total of $\boldsymbol{b_e}$**independent equations** for the $\boldsymbol{b_e}$ **unknown branch currents**
  - Since $\boldsymbol{n_e \leq n}$ and $\boldsymbol{b_e \leq b}$ it is better to use the essential nodes and essential branches to analyze the circuit (This results in fewer number of equations).
  - These hold for connected circuits. Disconnected circuits can be reduced to connected circuits before analysis.
  - Once all the equations are established, they can be solved using techniques from linear algebra.

Terminology

- An example:



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Writing the equations considering the essential nodes and branches:

- The circuits has four essential nodes (b, c, e and g) and six essential branches, denoted $i_1 - i_6$ for which the current is unknown.
- We derive three of the six simultaneous equations needed by applying KCL to any three of the four essential nodes. We use the nodes b, c and e to get:

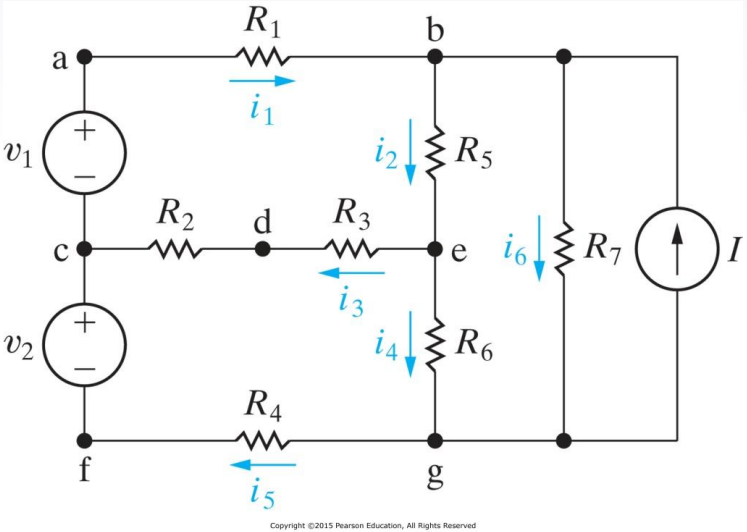
$$-i_1 + i_2 + i_6 - I = 0$$

$$i_1 - i_3 - i_5 = 0$$

$$i_3 + i_4 - i_2 = 0$$

1 Terminology

- An example:



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- We derive the remaining three equations by applying KVL around three meshes. Because the circuit has four meshes, we need to dismiss one mesh. We choose to dismiss $R_7 - I$ mesh, because we don't know the voltage across I. Using the other three meshes gives:

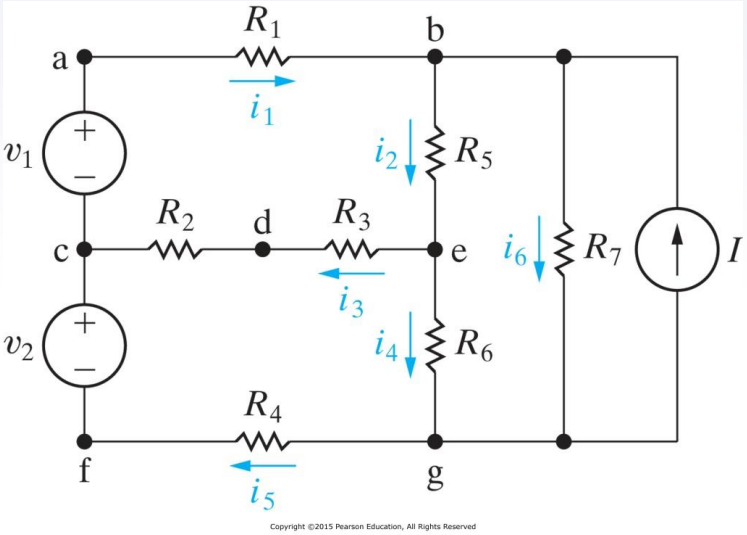
$$R_1 i_1 + R_5 i_2 + i_3(R_2 + R_3) - v_1 = 0$$

$$-i_3(R_2 + R_3) + R_6 i_4 + R_4 i_5 - v_2 = 0$$

$$-R_5 i_2 + R_7 i_6 - R_6 i_4 = 0$$

Terminology

- An example:



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- Rearranging the equations to facilitate their solution yields the set:

$$-i_1 + i_2 + 0i_3 + 0i_4 + 0i_5 + i_6 = I$$
$$i_1 + 0i_2 - i_3 + 0i_4 - i_5 + 0i_6 = 0$$
$$0i_1 - i_2 + i_3 + i_4 + 0i_5 + 0i_6 = 0$$

$$R_1 i_1 + R_5 i_2 + (R_2 + R_3) i_3 + 0i_4 + 0i_5 + 0i_6 = v_1$$
$$0i_1 + 0i_2 - (R_2 + R_3) i_3 + R_6 i_4 + R_4 i_5 + 0i_6 = v_2$$
$$0i_1 - R_5 i_2 + 0i_3 - R_6 i_4 + 0i_5 + R_7 i_6 = 0$$

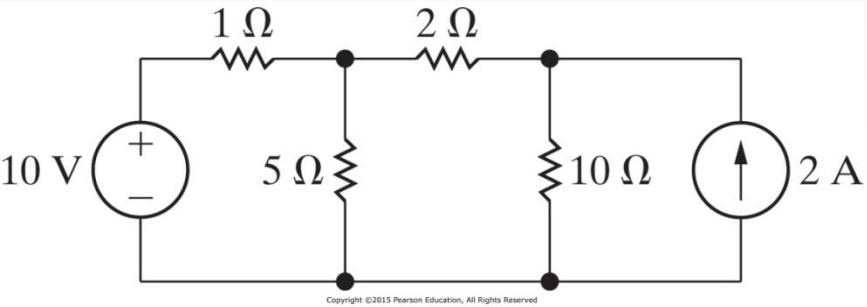
- By introducing new variables, we can describe a circuit with just $(\boldsymbol{n-1})$ equations or just $\mathbf{b} - (\boldsymbol{n-1})$ equations.
- The node-voltage method allows to describe a circuit with $(\boldsymbol{n_e - 1})$ equations.
- The mesh-current method allows to describe a circuit with $\boldsymbol{b_e - (n_e - 1)}$ equations.

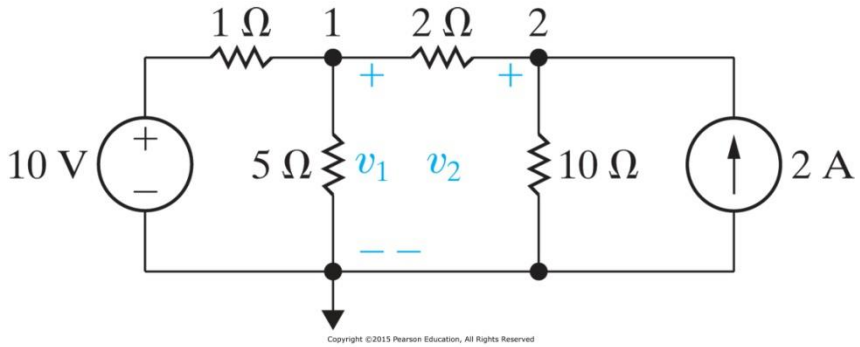
2 Introduction to the Node-Voltage method

- The node-voltage method is based on determining the voltages of the nodes in the circuit (relative to a chosen referene node).
- We use the essential nodes in the node-voltage method.
- Select one node as a reference node.
- Define the node voltages for the remaining nodes.
- Node voltages are defined as voltage rise from the reference node to the given node.
- Write down the node-voltage equations, i.e. Write down KCL equations at the nodes in terms of the defined node voltages.

2 Introduction to the Node-Voltage method

- An example

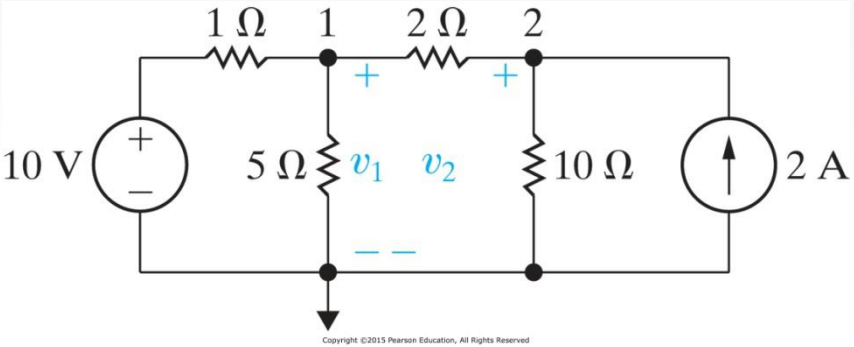




- The circuit has three essential nodes ($n_e = 3$). Therefore we need two ($n_e - 1$) node-voltage equations to describe the circuit.
- The next step is to select one of the three essential nodes as a reference node. Although the choice is arbitrary, the node with the most branches is usually a good choice (the lower node in the given circuit). We flag the chosen reference node with ⏚ symbol.
- Then two node voltages are defined as voltage rise from the reference node.

2 Introduction to the Node-Voltage method

- An example

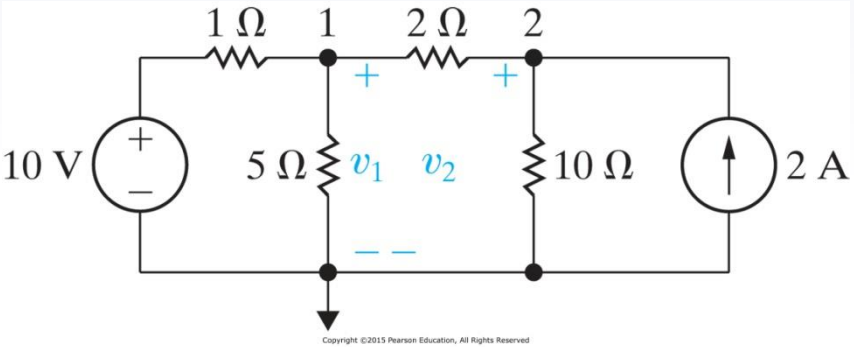


- We are now ready to generate the node-voltage equations.
- Using KCL, the currents are written in terms of voltages and resistances, considering every current away from a considered non-reference node through a resistor as voltage drop across the resistor:

- For node 1: $\frac{v_1 - 10}{1} + \frac{v_1}{5} + \frac{v_1 - v_2}{2} = 0$
- For node 2: $\frac{v_2 - v_1}{2} + \frac{v_2}{10} - 2 = 0$

2 Introduction to the Node-Voltage method

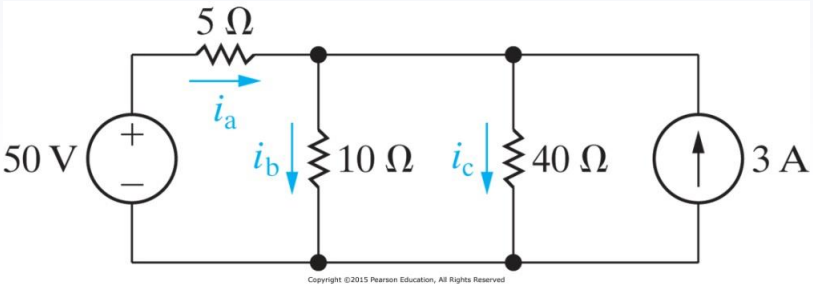
- An example



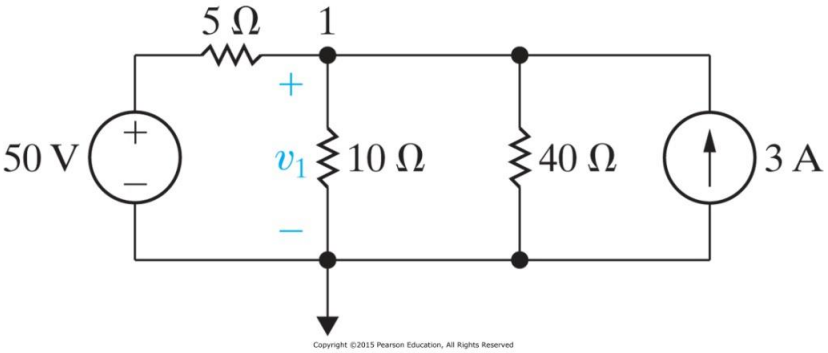
- Once the node voltages are known, all the brnch currents can be calculated.
- If there are n_e essential nodes in the circuit, $(n_e - 1)$ equations are sufficient to solve the circuit.
- The obtained node voltages are relative to the value of the reference voltage (i.e. the voltage at the reference node) – which is usually chosen as zero.

2 Introduction to the Node-Voltage method

- Another example



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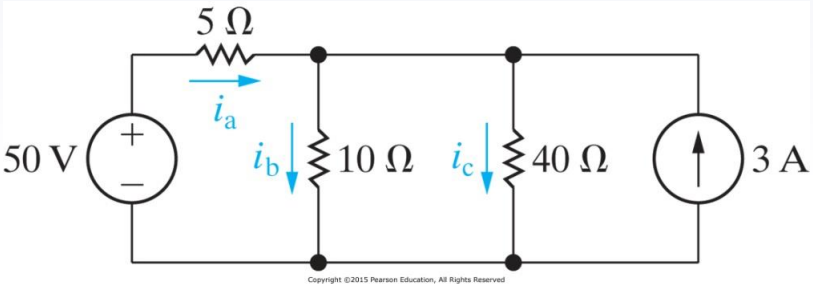
a) Use the node-voltage method of circuit analysis to find the branch currents i_a, i_b and i_c in the given circuit.

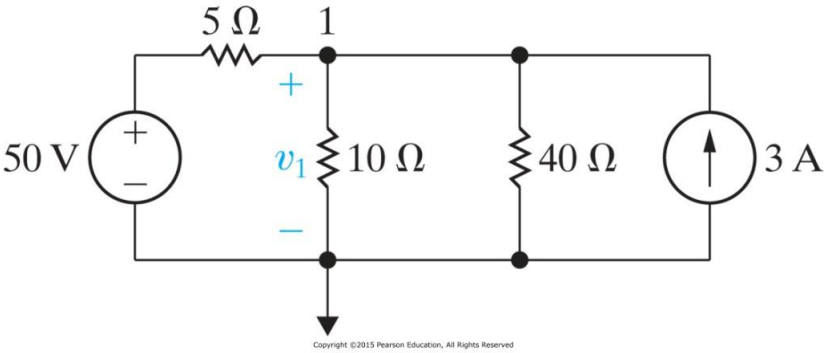
b) Find the power associated with each source and state whether the source is delivering or absorbing power.

Solution a) The circuit has two essential nodes; thus we need to write one node-voltage equation. We select the lower node as the reference node and define the unknown voltage as v_1.

2 Introduction to the Node-Voltage method

- Another example





Summing the currents away from node 1 generates the node-voltage equation:

$$\frac{v_1 - 50}{5} + \frac{v_1}{10} + \frac{v_1}{40} - 3 = 0$$

$$8v_1 - 400 + 4v_1 + v_1 = 120$$

$$13v_1 = 520$$

$$v_1 = 40V$$

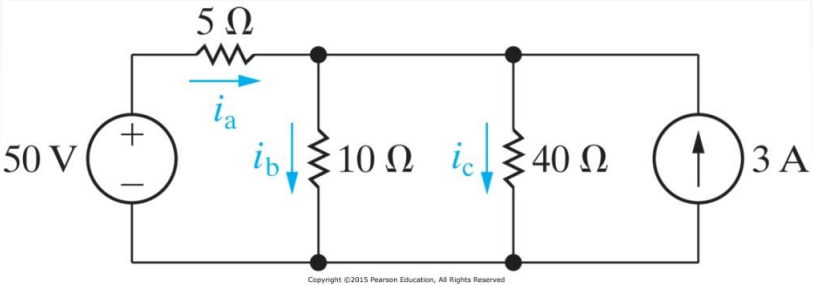
$$i_a = \frac{50 - 40}{5} = 2A$$

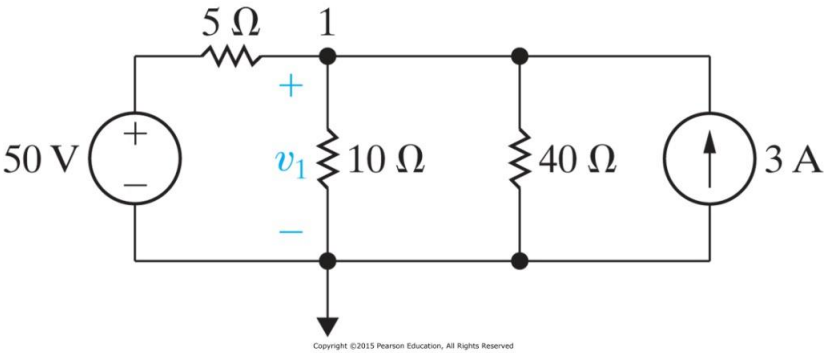
$$i_b = \frac{40}{10} = 4A$$

$$i_c = \frac{40}{40} = 1A$$

2 Introduction to the Node-Voltage method

- Another example





Solution b)

The power associated with the 50V source is:

$$p_{50V} = -50i_a = -100W \text{ (delivering)}$$

The power associated with the 3A source is:

$$p_{3A} = -3v_1 = -3(40) = -120W \text{ (delivering)}$$

Validation:

Considering that the total delivered power is 220W

The total power absorbed by the three resistors are

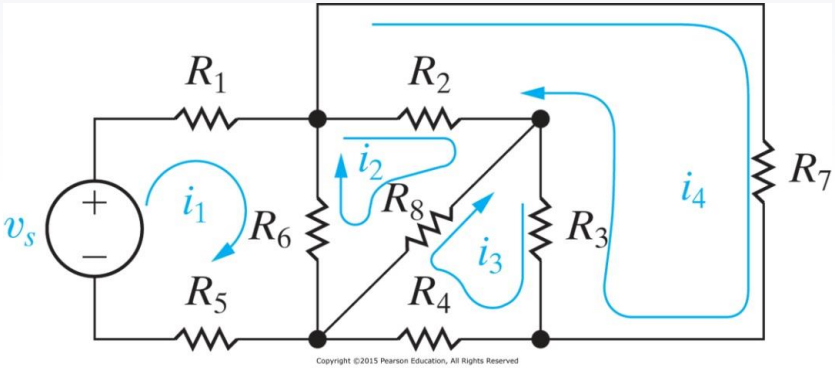
$$(2)^2 5 + (4)^2 10 + (1)^2 40 = 220W \text{ (absorbing)}$$

5 Introduction to the Mesh-Current Method

- The **mesh-current method** can be used in **planar circuits only**.
- If a circuit has $\boldsymbol{b_e}$ **essential branches** and $\boldsymbol{n_e}$ **essential nodes**, then $[b_e - (n_e - 1)]$ **mesh equations are needed** to solve the circuit.
- Mesh currents are **fictitious** currents which are assumed to exist **in the perimeter of every mesh.**
- **Different current** is assigned to **every mesh**.
- The currents in the branches are calculated **considering the currents of the meshes it belongs to** (Note that **more than one mesh current can flow through a circuit element or branch**).

5 Introduction to the Mesh-Current Method

- Example-1

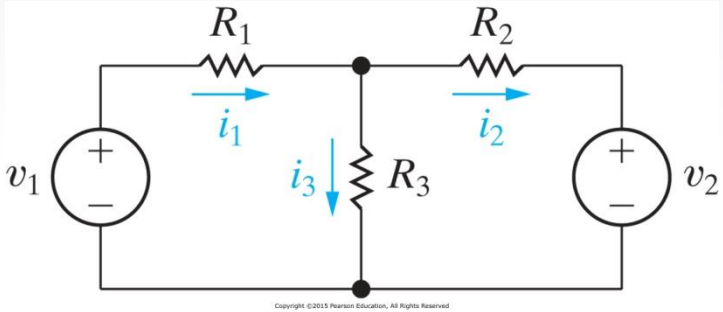


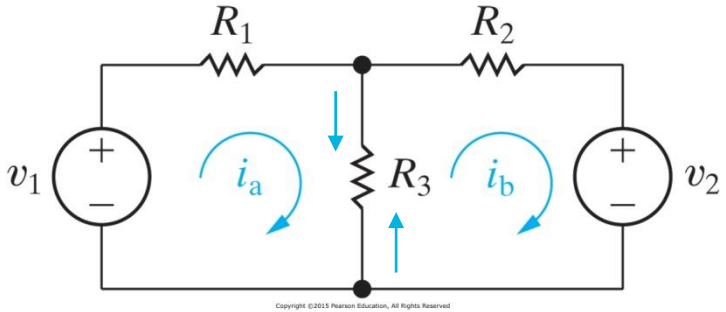
- There are **7 essential branches** where the current is unknown and **4 essential nodes**.
- Therefore, in order to solve it via the mesh-current method, we must write $[7-(4-1)]=4$ **mesh-current equations**.
- Mesh current automatically satisfies **Kirchoff's current law**. At any node in the circuit, a given mesh current **both enters and leaves the node**.

- Identifying **a mesh current in terms of a branch current** is not always possible. In the given circuit, $\boldsymbol{i_2}$ **is not equal to any brach current**. There is **no place** where an ampermeter can be inserted to measure the mesh current i_2. **On the other hand**, $\boldsymbol{i_1}$, $\boldsymbol{i_3}$ **and** $\boldsymbol{i_4}$ **can be identified with branch currents**.

Introduction to the Mesh-Current Method

- Example-2





- There are **2 essential nodes** and **3 essential branches**.
- Therefore we can write $[3-(2-1)]=2$ **voltage equations**.

$$-v_1 + i_a R_1 + (i_a - i_b)R_3 = 0$$
$$(i_b - i_a)R_3 + i_b R_2 + v_2 = 0$$

organizing both equations

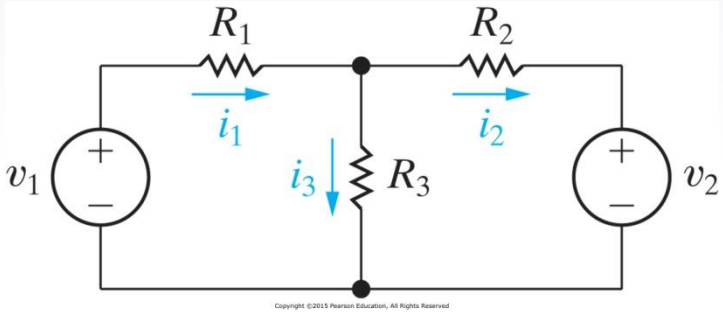
$$v_1 = i_a R_1 + (i_a - i_b)R_3$$
$$-v_2 = (i_b - i_a)R_3 + i_b R_2$$

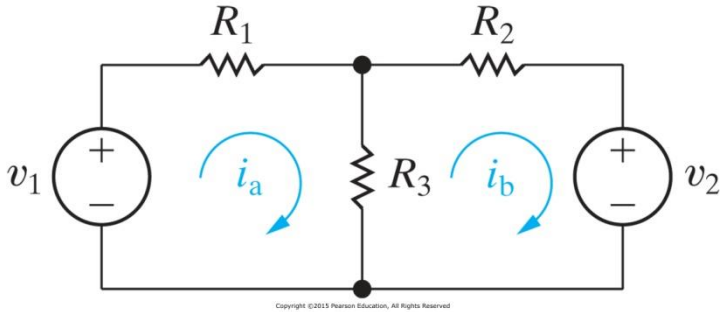
Collecting the coefficients of i_a and i_b

$$v_1 = i_a(R_1+R_3) - i_b R_3$$
$$-v_2 = -i_a R_3 + i_b(R_2+R_3)$$

5 Introduction to the Mesh-Current Method

- Example-2





$$v_1 = i_a(R_1+R_3) - i_bR_3$$
$$-v_2 = -i_aR_3 + i_b(R_2+R_3)$$

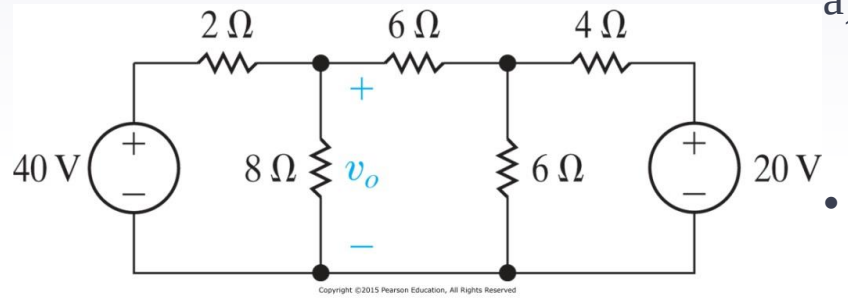
- **Branch currents can be expressed in terms of mesh currents**:

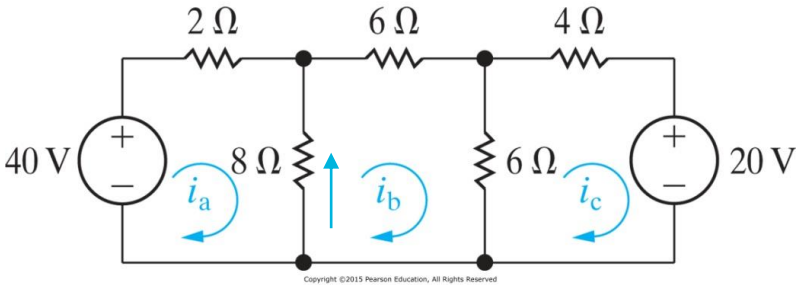
$$i_1 = i_a$$
$$i_2 = i_b$$
$$i_3 = i_a - i_b$$

- Once you **know the mesh currents**, you can easily **calculate the branch currents**. Once you **know the branch currrents**, you can **compute any voltage or powers of interest**.

5 Introduction to the Mesh-Current Method

- Example-3





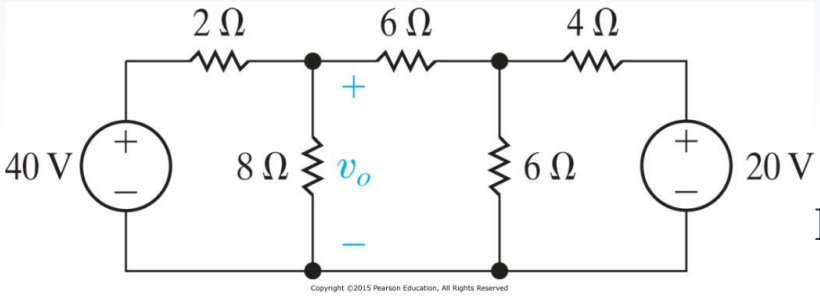
a) Use the **mesh-current method** to determine the **power associated with each voltage source** in the circuit.

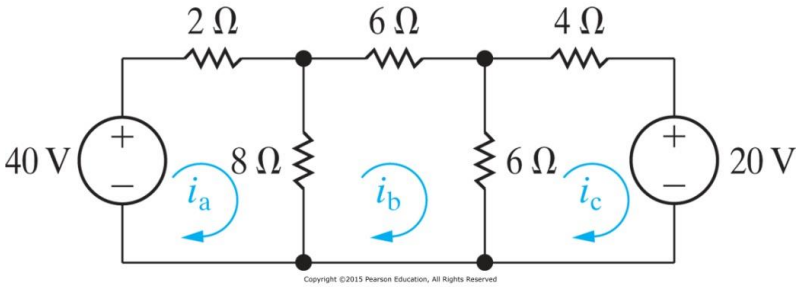
- There are **3 essential nodes** and **5 essential branches**. Therefore we must write $[5-(3-1)] = 3$ **mesh-current equations**.

$$-40 + 2i_a + 8(i_a - i_b) = 0$$
$$8(i_b - i_a) + 6i_b + 6(i_b - i_c) = 0$$
$$6(i_c - i_b) + 4i_c + 20 = 0$$

5 Introduction to the Mesh-Current Method

- Example-3





Organizing the equations

$$10i_a - 8i_b + 0i_c = 40$$

$$-8i_a + 20i_b - 6i_c = 0$$

$$0i_a - 6i_b + 10i_c = -20$$

Eliminating i_c from **equations 2 and 3**:

$$(Multiply\ with\ 5) \quad -8i_a + 20i_b - 6i_c = 0$$

$$(Multiply\ with\ 3) \quad 0i_a - 6i_b + 10i_c = -20$$

$$-40i_a + 100i_b - 30i_c = 0$$

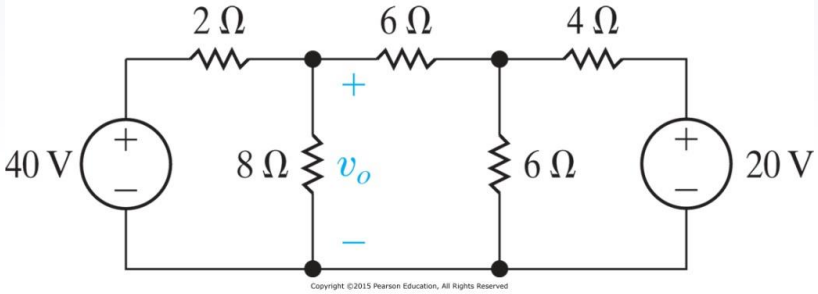
$$0i_a - 18i_b + 30i_c = -60$$

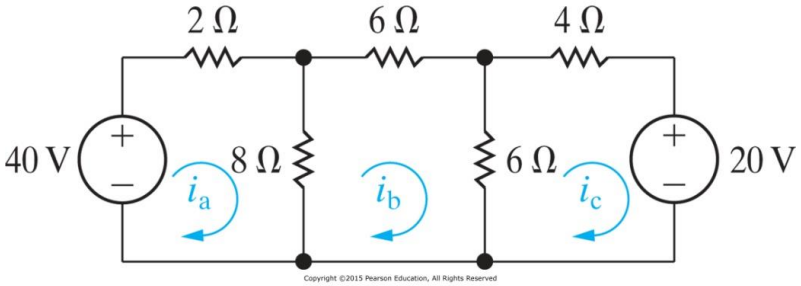
Adding the equations

$$-40i_a + 82i_b = -60$$

5 Introduction to the Mesh-Current Method

- Example-3





Using **the first original equation** and **the last derived equation** to **eliminate i_a,**

$$(Multiply\ with\ 4) \quad 10i_a - 8i_b = 40$$

$$-40i_a + 82i_b = -60$$

$$40i_a - 32i_b = 160$$

$$-40i_a + 82i_b = -60$$

We obtain

$$50i_b = 100$$

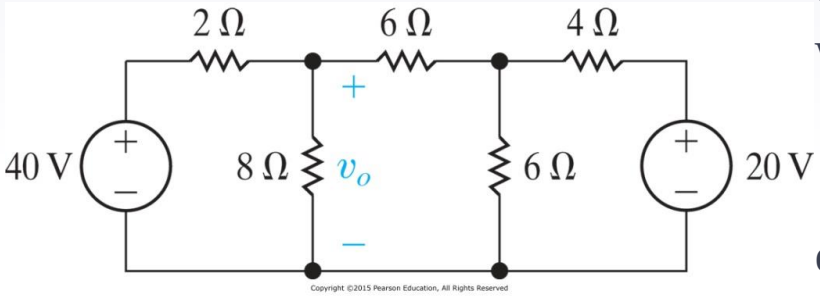
$$i_b = 2A$$

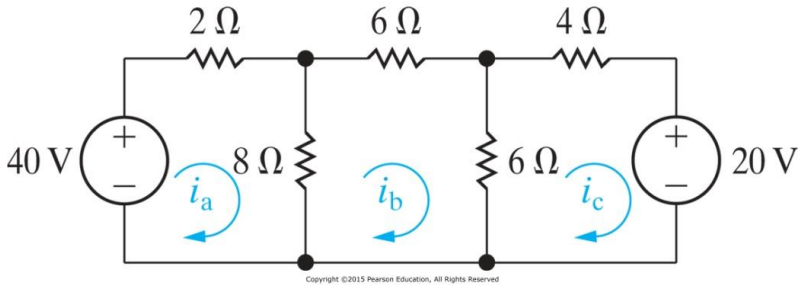
$$10i_a - 8(2) = 40 \qquad i_a = 5.6A$$

$$-6(2) + 10i_c = -20 \qquad i_c = -0.8A$$

5 Introduction to the Mesh-Current Method

- Example-3





The mesh current i_a **is identical with the branch current in the 40V source**, so the power associated with this source is

$$p_{40V} = -40i_a = -40(5.6) = -224W$$

The **minus sign means that this source is delivering power to the circuit**.

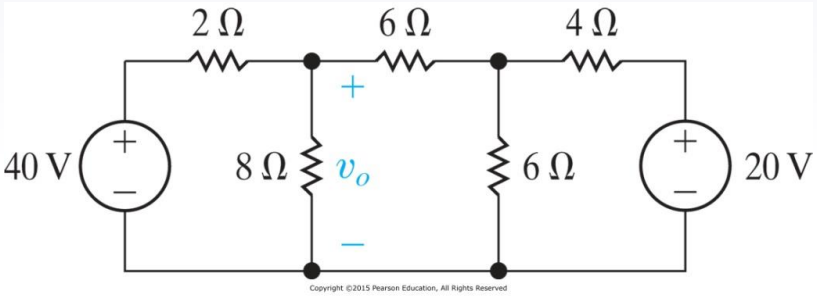
The mesh current i_c **is identical with the branch current in the 20V source**, so the power associated with this source is

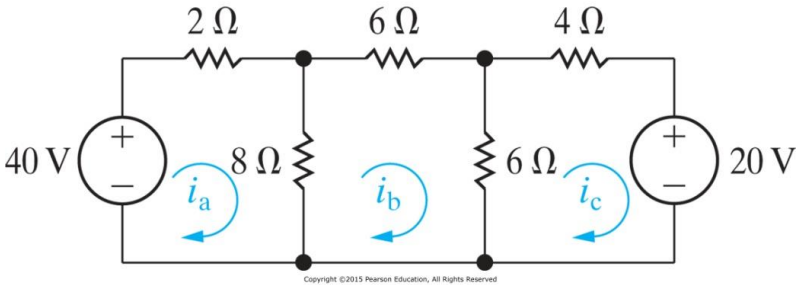
$$p_{20V} = 20i_c = 20(-0.8) = -16W$$

The 20 V source also **is delivering power to the network**.

5 Introduction to the Mesh-Current Method

- Example-3





b) Calculate **the voltage** v_o across the **8 Ω resistor**

The branch current in the 8 Ω resistor **in the direction of voltage drop** $\boldsymbol{v_o}$ is $\boldsymbol{i_a - i_b}$. Therefore

$$v_o = 8(i_a - i_b) = 8(5.6 - 2) = 8(3.6) = 28.8\ V$$

End of the Lesson (23.10.23)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist